

SELFRIDGE AFB, MI, USA

WMO: 725377

Lat: 42.608N

Lon: 82.818W

Elev: 580

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14804

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.4	(c) 5.4	(d) -6.4	(e) 4.0	(f) 2.4	(g) -1.2	(h) 5.3	(i) 7.2	(j) 25.5	(k) 24.1	(l) 23.5	(m) 25.2	(n) 8.4	(o) 230	(p) 0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	89.9	74.2	86.5	72.4	84.0	71.1	76.8	85.0	75.1	82.7	73.4	80.8	11.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	132.2	80.8	72.6	123.7	79.3	71.3	118.1	78.2	40.5	85.0	38.9	82.8	37.4	80.7	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	19.5	17.7	-6.0	94.8	6.7	2.7	-10.8	96.8	-14.7	98.4	-18.5	99.9	-23.3	101.9
			-6.4	80.3	6.7	2.4	-11.2	82.0	-15.1	83.4	-18.8	84.8	-23.6	86.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	24.3	27.6	34.5	46.3	58.0	68.0	72.2	70.9	64.4	52.6	40.8	31.0
	DBStd	18.87	11.10	10.84	10.32	8.72	8.16	7.08	5.71	5.35	7.39	8.11	8.63	9.64
	HDD50	3063	798	627	493	171	18	0	0	0	2	64	298	592
	HDD65	6409	1263	1046	946	566	248	48	7	11	100	393	726	1055
	CDD50	2818	1	1	12	58	266	540	688	648	434	146	21	3
	CDD65	689	0	0	1	3	32	138	230	194	82	9	0	0
	CDH74	5828	0	0	6	43	331	1284	2029	1456	621	58	0	0
	CDH80	1734	0	0	0	8	76	442	653	392	156	7	0	0
	WSAvg	8.6	10.3	9.9	10.0	9.8	8.2	7.2	7.0	6.7	7.0	8.4	9.5	9.3
	PrecAvg	33.4	2.2	1.9	2.4	3.2	3.6	3.0	3.3	3.2	3.1	2.4	2.5	2.4
(15) Precipitation	PrecMax	47.3	4.1	4.5	5.2	6.4	7.2	5.4	6.8	6.3	6.7	6.6	5.6	4.0
	PrecMin	23.4	0.6	0.3	0.9	0.7	1.5	1.1	0.7	0.6	0.5	0.5	0.7	0.9
	PrecStd	5.0	0.9	1.0	1.1	1.3	1.6	1.3	1.7	1.3	1.8	1.3	1.1	0.8
	PrecStd	5.0	0.9	1.0	1.1	1.3	1.6	1.3	1.7	1.3	1.8	1.3	1.1	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	61.1	72.1	79.6	86.0	92.2	93.3	91.4	89.8	80.6	66.4	58.9
		MCWB	52.8	53.4	57.7	63.2	69.0	74.3	76.4	76.4	73.8	69.2	58.4	55.0
	2%	DB	49.4	53.3	62.3	72.5	81.9	89.1	89.7	87.7	84.0	74.4	61.8	52.5
		MCWB	46.3	48.4	53.7	59.6	68.0	73.1	74.5	73.3	70.9	65.2	56.5	49.7
	5%	DB	42.6	46.2	55.2	66.4	77.7	85.7	86.3	84.1	81.1	70.3	58.2	47.8
		MCWB	39.9	41.9	48.7	55.8	65.8	71.5	72.7	72.3	68.9	62.9	53.6	44.6
	10%	DB	38.3	41.1	49.7	61.3	73.0	81.5	83.7	81.4	77.2	66.2	54.4	43.3
		MCWB	36.3	37.8	43.9	51.9	63.2	69.3	71.2	70.3	67.3	60.0	49.9	40.3
	0.4%	WB	53.8	55.4	59.7	66.6	73.4	78.0	80.0	78.2	75.9	71.4	61.3	55.4
		MCDB	54.9	57.9	67.9	77.1	80.0	87.2	88.7	87.6	84.2	77.7	63.4	57.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.4	48.5	54.8	61.8	70.7	75.2	77.2	76.1	73.5	67.9	57.7	50.4
		MCDB	47.8	52.3	60.7	69.2	78.4	85.4	85.3	83.3	80.9	71.7	60.6	52.4
	5%	WB	40.5	42.1	49.9	57.2	68.0	73.1	75.4	74.3	71.3	64.1	54.3	45.1
		MCDB	42.2	45.3	55.6	64.5	75.3	82.0	82.5	81.0	77.4	68.7	57.4	47.5
	10%	WB	36.5	38.2	44.3	53.0	64.8	71.2	73.4	72.6	69.3	61.0	50.2	40.8
		MCDB	38.3	40.9	48.0	60.0	71.2	79.3	80.5	79.2	74.3	65.1	53.9	43.2
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	15.6	17.1	19.7	20.6	20.1	19.5	18.3	19.9	17.7	14.1	12.3
		MCDBR	17.4	21.2	26.7	28.7	25.4	24.1	21.3	21.0	22.8	21.9	18.6	17.4
		MCWBR	15.5	16.8	18.3	18.3	14.6	11.6	10.0	10.3	11.4	13.7	14.2	14.7
	5% WB	MCDBR	16.5	20.3	25.2	24.9	22.8	21.3	18.9	18.4	19.5	19.0	16.9	16.5
		MCWBR	15.5	17.2	18.2	17.6	15.1	11.7	10.3	10.1	11.2	13.1	14.6	15.0
(40) Clear-Sky Solar Irradiance	taub		0.306	0.333	0.350	0.396	0.424	0.444	0.433	0.423	0.387	0.360	0.321	0.304
	taud		2.374	2.283	2.354	2.246	2.221	2.202	2.256	2.273	2.356	2.415	2.486	2.430
	Ebn at Noon		263	272	283	276	271	265	266	265	267	261	257	254
	Edh at Noon		25	33	35	42	44	45	42	40	35	29	23	22
	RadAvg		449	729	1111	1466	1717	1945	1932	1674	1348	872	535	379
(45) All-Sky Solar Radiation	RadStd		52	78	112	128	147	117	96	69	117	75	68	31

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page